

Title: Task Scheduler Optimization

You are given a list of tasks represented by characters (A-Z), and a non-negative integer n representing a cooldown period. Each task takes 1 unit of time. Two same tasks must have at least n units of time between them.

Return the minimum time required to complete all tasks.

Example:

Input: tasks = ["A","A","A","B","B","B"], $n = 2$

Output: 8

Constraints:

- $1 \leq \text{tasks.length} \leq 10^4$
- tasks[i] is an uppercase English letter
- $0 \leq n \leq 100$